

A Parsimonious Code: Allophonic Voicing and the Limits of Phonological Supervision in Romanised Tamil and Malayalam Transliteration

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Abstract

Tamil orthography does not dedicate separate graphemes to voiced and voiceless stops: a single graphemic stop series spans $[k]$ and $[g]$, with phonetic realisation systematically conditioned by phonotactic environment. Malayalam retains the full Indic voiced series but confines it largely to the Sanskrit stratum, writing the voiceless grapheme in native vocabulary under the same phonotactic conditioning. In Indic natural language processing, this trait has frequently been misdiagnosed as an “orthographic underspecification” deficit, motivating interventions that inject explicit phonological priors into neural sequence models. This paper evaluates this hypothesis empirically and theoretically, finding that native orthography is in fact an information-theoretically parsimonious code: an unconditional voiceless baseline achieves 79.49% accuracy in Tamil and 89.77% in Malayalam across all phonotactic contexts. Measuring stop voicing over 150,000 benchmark word pairs per language confirms that phonotactic context is genuinely informative (mutual information $I(V; C) = 0.091$ bits in Tamil, 0.106 in Malayalam). However, because the voiceless spelling remains the plurality outcome in every phonotactic slot in crowdsourced benchmarks ($P(\text{voiced}) < 0.50$ in all contexts, with Malayalam post-nasal touching the boundary at 0.492), conditioning on context alters the argmax decision boundary by 0.00%, a property formalised here as *argmax invariance*. Parsimony here is decision-theoretic rather than entropic: the omitted distinction carries zero Bayes risk under 0–1 loss, not negligible conditional entropy ($H(V | C) = 0.641$ bits in Tamil, 0.370 in Malayalam), whose residual is traced below to annotator transcription variance rather than to native phonological structure. Across controlled en $\rightarrow$ indic transliteration scaling experiments spanning 25k to 1.0M training pairs and two distinct architectures, auxiliary phonological supervision yields no measurable gain at low resource scales (Malayalam exact match +0.15 points, Tamil -0.14 , across three random seeds at 25k pairs; paired McNemar $p = 0.90$ and 0.59) and reverses to significant degradation at 1.0M pairs (-2.33% on Tamil, $p = 0.0003$). Residual target entropy in indic $\rightarrow$ en benchmarks traces to human transcription conventions: independent annotators disagree on 69.26% of Tamil post-nasal plosive slots on identical words in Dakshina (44.50% pairwise). In contrast, conversational Tanglish from YouTube comments exhibits an empirical 64.00% voiced spelling rate post-nasally. Finally, this work provides actionable engineering blueprints for handling this duality in practical input methods (en $\rightarrow$ indic) and controlled generation (indic $\rightarrow$ en), and releases the open VALIMELI-SPEECH index (5,325 phonotactically annotated native spoken Tamil audio recordings across 7 contributors, 96.0% from one speaker, spanning 12,435 evaluated plosive slots across 11,590 core syllable positions) under CC BY-SA 4.0.

1 Introduction

Machine transliteration between the Latin alphabet and the indigenous Brahmic scripts of South Asia is a foundational component of keyboard input method engines (IMEs), search indexing, named-entity recognition,

and conversational text normalisation Madhani et al. [2023], Roark et al. [2020], Ramesh et al. [2022]. While sequence-to-sequence models frequently exceed 60% exact match on held-out benchmarks for multi-stop scripts like Hindi, Telugu, and Kannada, performance on Tamil and Malayalam is historically lower.

A recurring explanation in the computational literature attributes this disparity to *orthographic underspecification*. Tamil orthography maintains an autonomous single graphemic stop series (*vallinam* / hard plosives in the classical grammatical tradition, Niklas 1988: *ka, ca, ʔa, ta, pa, ʔa*) that maps to multiple phonetic realisations: Tamil writes $[k]$, $[g]$, $[\gamma]$, and $[x]$ with the single grapheme *ka*. Malayalam retains the full Brahmic four-way plosive inventory (*ka, kha, ga, gha*) for Indo-Aryan borrowings, but preserves ancestral Dravidian complementary distribution in native vocabulary, writing voiceless graphemes for native intervocalic and post-nasal stops. It has thus been widely assumed that neural models face a fundamental information bottleneck because Dravidian orthographies omit surface allophonic voicing distinctions present in other scripts.

Principled Orthographic Parsimony. Classical phonological theory refutes the deficit hypothesis. Under the classical *Phonemic Principle* Swadesh [1934], Caldwell [1875], Zvelebil [1970], Ramprashanth et al. [2025], writing systems encode contrastive phonemes rather than predictable surface allophones. In the core native Dravidian lexicon, stop voicing exists in strict *complementary distribution*:

1. **Word-initial** (#_): strictly voiceless fortis ($[k, t, p]$, e.g., *pakkam* $\rightarrow [p]a[kk]am$).
2. **Post-nasal** (N_): obligatorily voiced via nasal assimilation / *puṇarcci* (morphophonemic *sandhi*; e.g., *tampi* / *thambi* $\rightarrow [b]$, *pantu* / *pandhu* $\rightarrow [d]$).
3. **Intervocalic** (V_V): voiced or spirantised lenis (*paṭam* / *padam* $\rightarrow [d]$, *alaku* / *azhagu* $\rightarrow [g]$).

Because voicing is completely predictable from local phonotactics, dedicating separate graphemes to $[k]$ and $[g]$ in native vocabulary would constitute redundant functional overhead. Native orthography is therefore a parsimonious code.

Acoustic Realisation in Native Speech. To illustrate that complementary voicing is a physically grounded property of spoken speech rather than a theoretical abstraction, acoustic waveforms and Short-Time Fourier Transform (STFT) spectrograms from native speaker recordings in the Wikimedia Commons / Lingua Libre repository¹ were examined (Figure 1 and Table 1). As illustrated in Figure 1, word-initial plosives (*paṭam* / *padam* $[p-]$) and geminates exhibit an absence of low-frequency periodic energy followed by silent closure gaps in geminates (*pakkam* $[pak : am]$). In stark contrast, post-nasal plosives (*tampi* / *thambi* $[tambi]$, *maruntu* / *marundhu* $[marundu]$) and intervocalic plosives exhibit continuous fundamental voicing energy bridging the nasal and plosive segments. Across the full corpus of 5,325 recordings (12,435 plosive slots evaluated across 11,590 core syllable positions and 845 post-consonantal and coda slots), physical voicing is strongly conditioned by phonotactic environment: word-initial plosives are 99.72% voiceless (0.0279 bits entropy; mean closure 44.0 ms), post-nasal plosives are 85.14% voiced (78.8% voice bar ratio; mean closure 4.7 ms), and intervocalic plosives exhibit 56.77% lenis voicing. Overall, local phonotactics correctly predicts physical spoken voicing across 70.29% of all consonant tokens.

To support future research in Dravidian phonetics, text-to-speech (TTS), speech recognition (ASR), and phonetic typing IMEs, this paper curates and releases the VALIMELI-SPEECH index²: an open, phonotactically annotated index of 5,325 native spoken Tamil audio recordings (12,435 plosive slots systematically

¹Audio recordings sourced from the Lingua Libre project on Wikimedia Commons, contributed by native speakers (User: Sriveenkat, User: Manimaran96, User: StarlitNocturne) under the Creative Commons Attribution-ShareAlike 4.0 International license (CC BY-SA 4.0).

²The VALIMELI-SPEECH index, extracted acoustic features, and replication code are openly available at <https://github.com/oligoglot/valimeli>.

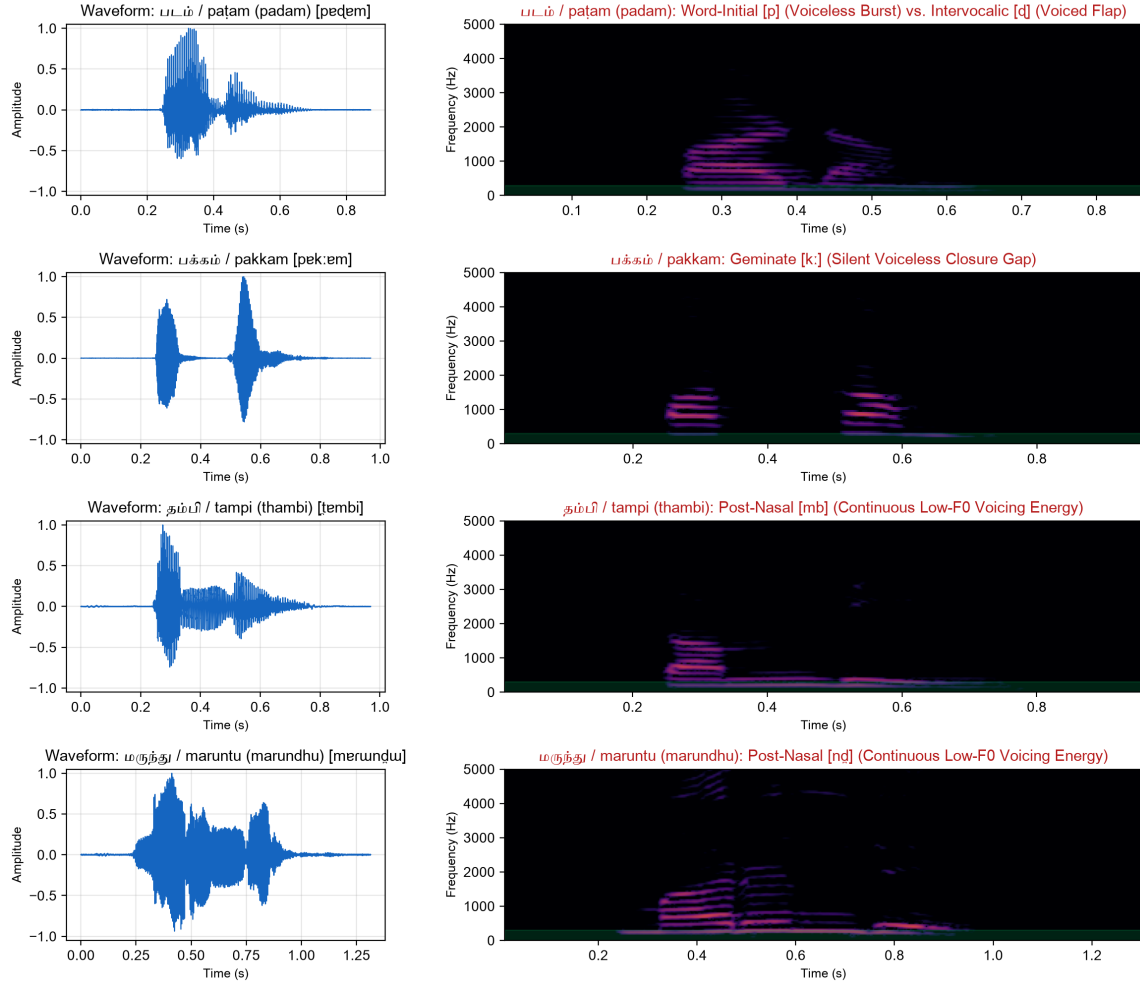


Figure 1: Acoustic waveforms and STFT spectrograms of native Tamil speech recordings (Lingua Libre corpus). Voicing is physically evident in the continuous low-frequency periodic energy during post-nasal (*tampi / thambi* [mb], *maruntu / marundhu* [nd]) and intervocalic contexts (*paṭam / paṭam* [d]), contrasted with the silent closure gap in geminates (*pakkam* [k:]). Panels indicate native Tamil script, standard ISO 15919 romanisation, colloquial spelling, and phonetic IPA.

categorised into word-initial, geminate, post-nasal, intervocalic, and coda environments, across 7 contributors, with 96.0% contributed by primary native speaker *User: Sriveenkat*) with full speaker attributions and direct audio stream URLs under CC BY-SA 4.0.

The Unassimilated Loanword Stratum. A subtle complication arises from modern borrowings (English, Sanskrit, Hindustani). While classical Tamil phonology nativised loanwords to conform to native allophony,³ modern 20th-century borrowings preserve foreign voiced stops in speech (*Gandhi* → *kānti* [g–], *Dilli* → *tilli* [d–], *Bus* → *pas* [b–]). Malayalam accommodates this stratum orthographically through the Brahmic Grantha plosive series (*ga*, *da*, *ba*), reserving those graphemes for foreign voicing. Tamil, whose stop graphemes stand

³Under classical nativisation rules (*Tolkāppiyam*, *Collatikāram sūtra* 397; Niklas 1988), loanwords were adapted to native phonotactics by devoicing non-native voiced stops in initial position (e.g., Sanskrit *danta* → Tamil *tantam* [t–], *gaṇa* → *kaṇam* [k–]). In contrast, modern borrowings preserve voiced stops in speech while using native single-series graphemes in writing, creating orthographic duality in crowdsourced romanisation.

Tamil Word	<i>Eḷuttu</i> Unit	Context	Phonetic IPA	Observed Acoustic Realisation	Roman Spellings
<i>paṭam</i> (<i>padam</i>)	<i>pa</i>	Word-Initial (#_)	[p]	Voiceless burst / no low F_0	p
<i>paṭam</i> (<i>padam</i>)	<i>ṭa</i>	Intervocalic (V_V)	[d]	Voiced flap periodicity	d / t
<i>pakkam</i>	<i>kka</i>	Geminate (C_C)	[k:]	Voiceless silent closure gap	kk
<i>tampi</i> (<i>thambi</i>)	<i>mpi</i>	Post-Nasal (N_)	[mb]	Continuous low- F_0 voicing bar	mb / mp
<i>makalīn</i> (<i>magalin</i>)	<i>ka</i>	Intervocalic (V_V)	[ɣ]	Continuous velar voicing periodicity	g / k
<i>kalyāṇaikaḷōṭu</i>	<i>nka</i>	Post-Nasal (N_)	[ŋg]	Continuous velar voicing bar	ng / ngg

Table 1: Empirical phonetic and acoustic alignment of Tamil stop graphemes (*vallinam*) from verified Wikimedia Commons speech recordings (Lingua Libre corpus). Across the full corpus of 5,325 recordings (12,435 plosive slots evaluated across 11,590 core syllable positions and 845 post-consonantal and coda slots; §1, Table 2), post-nasals exhibit 85.14% glottal voicing continuity (78.8% voice bar ratio, 4.7 ms mean closure), while word-initial plosives are 99.72% voiceless (0.0279 bits entropy). Voicing is realised in spoken acoustics as systematically conditioned by *eḷuttu* phonotactic position in native vocabulary.

in one-to-one correspondence with its contrastive stop phonemes, writes the same borrowings with the native series. The residual voicing signal in modern multi-reference corpora is therefore lexical and target-side in origin: unassimilated borrowings carrying source-language voicing yield an empirical 9.79% word-initial voiced romanisation rate (Table 3) and a 10.27% split-slot disagreement rate (Table 4).

Contributions. The empirical evaluation examines phonological inductive bias across data scales from 25k to 1.0M pairs:

1. **Argmax Invariance** (§3): Information-theoretic analysis establishes that $P(\text{voiced} \mid C) < 0.50$ in every phonotactic context in benchmark data. Context-conditioned argmax rules achieve 0.00% decision error reduction over an unconditional voiceless default.
2. **Multi-Seed Scaling Dynamics** (§5): Across multi-seed runs (seeds 42, 43, 44), phonological supervision yields overlapping performance at 25k low-resource pairs ($\Delta \leq 0.15$ points, $p \geq 0.58$), decays across intermediate scales, and reverses to significant degradation at 1.0M (-2.33% , $p = 0.0003$).
3. **Target-Side Duality** (§4): Empirical auditing demonstrates that residual entropy sits in target-side romanisation conventions (69.26% disagreement on post-nasal plosive slots in Dakshina, 44.50% pairwise), while in-the-wild Tanglish from YouTube comments exhibits a 64.00% voiced spelling rate.
4. **Open Speech Index & Acoustic Grounding** (§1, §3, §7): This study curates and releases the VALIMELI-SPEECH index (5,325 phonotactically annotated native spoken Tamil audio recordings across 7 contributors, with 96.0% from one primary speaker, spanning 12,435 plosive slots under CC BY-SA 4.0), contrasts physical acoustic voicing entropy with benchmark text entropy (+32.92% argmax error reduction in speech vs. 0.00% in text), and provides architectural blueprints for IMEs, transliteration, and Dravidian ASR.

2 Related Work

The Deficit Hypothesis in Dravidian Transliteration. Machine transliteration benchmarks across South Asian languages have consistently observed a marked performance disparity between scripts with full phonetic stop inventories and those with single-series stops. In the foundational AKSHARANTAR and INDICXLIT benchmark Madhani et al. [2023], sequence-to-sequence models trained across Indic languages systematically achieved lower exact-match accuracy on Dravidian languages with single-stop inventories (such as Tamil and

Malayalam) compared to multi-stop scripts (such as Hindi or Marathi). In our own controlled sequence-to-sequence baseline replications on these benchmarks (§5, Table 5), standard character models achieve only 58.19% exact-match accuracy for Tamil and 51.08% for Malayalam at the 250k training scale. In natural language processing discussions, this performance gap has frequently been framed around what we term the *orthographic underspecification deficit* hypothesis: the assumption that because the native orthographies collapse voiced and voiceless phones into a single graphemic series, the resulting script ambiguity acts as an information bottleneck in text generation. Madhani et al. [2023] noted that Dravidian languages exhibit substantial alignment ambiguity due to one-to-many and many-to-one character divergences, while Roark et al. [2020] demonstrated that in the DAKSHINA dataset, human annotators exhibit widespread spelling duality when transcribing Dravidian texts into the Latin script, reflecting uncertainty over whether to transcribe native phonemes or surface allophones. Such observations have motivated recurring proposals to inject explicit phonological features, phonetic lookup tables, or multi-task auxiliary loss heads into neural architectures.

Classical and Instrumental Dravidian Phonology. In contrast to the computational deficit narrative, classical Dravidian grammatical and phonetic theory treats the single stop series as a parsimonious design. The earliest extant Tamil grammatical treatise, *Tolkāppiyam* (*Eluttatikāram*, Niklas 1988), establishes a rigorous structural taxonomy: consonants (*meyyeḷuttu*) are partitioned into *vallinam* (hard plosives), *mellinam* (soft nasals), and *iṭaiyinam* (medial sonorants). The phonetic realisation of plosives is treated in classical grammar as conditioned by morphophonemic juncture (*puṇarcci / sandhi*). Caldwell [1875] formulated the classical comparative Dravidian “law of the convertibility of surds and sonants,” demonstrating that Proto-Dravidian occlusives were uniformly voiceless initially and when geminated, but obligatorily acquired voicing post-nasally and lenition intervocalically Zvelebil [1970]. Instrumental phonetic investigations Lisker [1958], Christdas [1988], Keane [2004] confirmed that native Tamil stops lack phonemic voicing contrasts; stop voicing is entirely allophonic and strictly conditioned by local phonotactic environment. Ramprashanth et al. [2025] analysed the mathematical distributions of biconsonantal clusters in Old Tamil Sangam literature, demonstrating that geminate (C_1C_1) and post-nasal (NC_1) cluster architectures have remained phonotactically invariant for two millennia.

Inductive Biases and Tokenisation in Neural Models. The injection of structural inductive biases into neural language models has yielded conflicting results across morphology and phonology. Recently, Vemula et al. [2025] investigated tokenisation strategies across morphologically rich Indic languages, concluding that unigram language models dominate byte-pair encoding (BPE) and demonstrating that explicit morphological alignment in tokenisers fails to provide consistent advantages over unconstrained subword segmentations. This paper extends this empirical scrutiny to phonological inductive biases. As demonstrated in §5, while phonology-aware string tags strictly outperform morphology-aware tokenisation across all four evaluation partitions (+0.35 to +2.48 points), neither inductive bias overcomes the fundamental information-theoretic barrier imposed by target-side reference variance.

3 Voicing Entropy and Argmax Invariance

Setup. The information-theoretic audit evaluates 150,000 word pairs per language from the Dakshina training split Roark et al. [2020] in Aksharantar Madhani et al. [2023]. Each native word is segmented into *eḷuttu* units (syllabic graphemic units / *aḱṣaras*, Niklas 1988) and aligned to its Latin transcription with a dynamic-programming aligner over candidate transcriptions. The aligner categorises every aligned plosive into five phonotactic contexts: word-initial, geminate, post-consonantal, intervocalic, and post-nasal.

Corpus / Modality	Analysed units	Coverage %	Plosive slots	$H(V)$ bits	$H(V C)$ bits	$I(V; C)$ bits	H red. %	Argmax gain %
Tamil (Dakshina text)	127,626	85.08	367,507	0.7319	0.6410	0.0909	12.42	0.00
Malayalam (Dakshina text)	100,716	67.14	234,476	0.4761	0.3699	0.1062	22.30	0.00
Tamil (Spoken acoustics)	5,325	100.0	12,435	0.9906	0.7555	0.2351	23.73	+32.92

Table 2: Voicing entropy of plosives across written orthography (Dakshina) and physical spoken speech acoustics (VALIMELI-SPEECH / Lingua Libre). Coverage for the spoken modality reflects 100.0% evaluation across the 5,325 waveforms in the VALIMELI-SPEECH index (12,435 plosive slots). While phonotactic context provides mutual information in both modalities ($I(V; C) > 0$), argmax decision gain is exactly **0.00%** in text benchmarks because Latin crowdsourcing maintains a voiceless plurality ($P(\text{voiced}) < 0.50$) across all environments. In spoken speech acoustics, phonotactic context actively shifts the decision boundary in post-nasal (85.14% voiced) and intervocalic (56.77% voiced) positions, achieving a **+32.92%** relative error reduction (from 44.29% error down to 29.71%; 70.29% rule accuracy). Malayalam native-stratum alignment (67.14%) isolates indigenous phonology; unaligned pairs are strongly enriched in Sanskrit/Grantha loan consonants ($7.30\times$ higher odds, $p < 10^{-300}$; §8).

Context is informative. Table 2 reports the information-theoretic quantities. Unconditional voicing entropy is 0.7319 bits in Tamil and 0.4761 in Malayalam; conditioning on phonotactic context reduces it to 0.6410 and 0.3699 bits, yielding mutual information of $I(V; C) = 0.0909$ bits (12.42% reduction; bootstrap 95% CI [0.0897, 0.0921]) in Tamil and 0.1062 bits (22.30% reduction; bootstrap 95% CI [0.1047, 0.1079]) in Malayalam. Sensitivity testing confirms this is robust to edit tolerance: tightening the aligner to exact match ($d = 0$) yields $I(V; C) = 0.1114$ bits (Tamil, 54.3% coverage) and 0.1043 bits (Malayalam, 35.9% coverage). Context is undeniably informative.

The Argmax Never Moves. Table 3 reveals why this mutual information fails to alter decision outcomes. In Tamil, word-initial plosives are romanised as voiced 9.79% of the time ($n = 62,194$), whereas post-nasal plosives are voiced 47.03% of the time ($n = 33,969$). Yet $0.4703 < 0.50$. In the exact context where spoken phonology mandates voicing, voiceless romanisations remain the plurality in crowdsourced benchmarks. In Malayalam, native stops exhibit a similar distribution (0.06% initial over $n = 32,323$ vs. 49.20% post-nasal over $n = 15,132$). While the Malayalam post-nasal point estimate strictly preserves the voiceless argmax ($0.4920 < 0.50$), its bootstrap 95% confidence interval ([48.40%, 50.00%]) touches the decision boundary within sampling error, indicating an empirical tie between transcription conventions in crowdsourced text. Consequently, an optimal context-conditional classifier and an unconditional “always voiceless” classifier produce identical predictions. Moving to context-conditioned rules reduces argmax decision error by exactly **0.00%** in both languages. In Malayalam, native-stratum alignment coverage reaches 67.14% (100,716 pairs; 70.89% plosive coverage); an empirical contingency audit confirms that the 32.86% unaligned stratum (49,284 pairs) is strongly enriched in Sanskrit/Grantha orthographic markers (84.49% contain loanword graphemes vs. 42.73% in aligned pairs; odds ratio $7.30\times$, $\chi^2 = 24,203.8$, $p < 10^{-300}$), while unaligned pairs actually exhibit slightly fewer complex conjunct ligatures (46.17% vs. 51.01%). This establishes that unaligned data reflects non-native phonological borrowings rather than structural limitations of native Dravidian phonology. This property is formalised as follows:

Proposition 1 (Argmax Invariance under Skewed Allophony). *Let $\mathcal{V} = \{0, 1\}$ denote binary stop voicing (0 = voiceless, 1 = voiced) and let $\mathcal{C}$ denote the finite set of phonotactic environments. If there exists $v^* \in \mathcal{V}$ such that for all $c \in \mathcal{C}$,*

$$v^* \in \arg \max_{v \in \mathcal{V}} P(V = v \mid C = c), \quad (1)$$

Context	Tamil (Text)			Tamil (Spoken Speech)			Malayalam (Text)		
	n	$P(v)$	H	n	$P(v)$	H	n	$P(v)$	H
Word-initial (#_)	62,194	0.0979	0.4623	2,131	0.0028	0.0279	32,323	0.0006	0.0072
Geminate (C_C)	63,786	0.0055	0.0493	2,386	0.3022	0.8839	51,987	0.0002	0.0027
Post-consonantal / Coda	23,062	0.2067	0.7352	845	0.4000	0.9707	6,644	0.0045	0.0417
Intervocalic (V_V)	184,496	0.2611	0.8285	5,566	0.5677	0.9867	128,390	0.1283	0.5527
Post-nasal (N_)	33,969	0.4703	0.9975	1,507	0.8514	0.6064	15,132	0.4920	0.9998

Table 3: Voicing probability $P(v) = P(\text{voiced})$ and entropy H (bits) across phonotactic environments in text corpora (Dakshina) versus physical speech acoustics (VALIMELI-SPEECH / Lingua Libre). In written text, $P(v) < 0.50$ in every single cell, so the argmax remains unconditionally voiceless; for Malayalam post-nasal slots ($P(v) = 0.4920$), bootstrap 95% CI [48.40%, 50.00%] touches the decision boundary within sampling error. In spoken speech acoustics, $P(v)$ flips decisively in post-nasal (85.14% voiced) and intervocalic (56.77% voiced) environments, while word-initial plosives are virtually noiseless (99.72% voiceless, $H = 0.0279$ bits). Geminate low- F_0 energy reflects vowel formant bleed into the 50.5 ms silent occlusion gap (§8). Malayalam native alignment (67.14%) isolates indigenous phonology; unaligned pairs are strongly enriched in Sanskrit/Grantha loan markers ($7.30\times$ higher odds, $p < 10^{-300}$; §8).

then $v^* \in \arg \max_{v \in \mathcal{V}} P(V = v)$, and the Bayes classification risk under context conditioning $R(\hat{v}_C)$ equals the unconditional Bayes risk $R(\hat{v}_\emptyset)$ under 0 – 1 loss $\ell(\hat{v}, v) = \mathbb{I}(\hat{v} \neq v)$:

$$R(\hat{v}_C) = \sum_{c \in \mathcal{C}} P(C = c) \left(1 - \max_{v \in \mathcal{V}} P(V = v \mid C = c) \right) = 1 - P(V = v^*) = R(\hat{v}_\emptyset). \quad (2)$$

Consequently, the Bayes error reduction is identically zero ($\Delta R \equiv R(\hat{v}_\emptyset) - R(\hat{v}_C) = 0$). This holds irrespective of the mutual information $I(V; C)$, and in particular when $I(V; C) > 0$, which is the empirically relevant case here (Table 2).

Proof. Because $v^* \in \arg \max_v P(V = v \mid C = c)$ for all $c \in \mathcal{C}$, the conditional maximum is $\max_v P(V = v \mid C = c) = P(V = v^* \mid C = c)$. Summing over context marginals yields $\sum_{c \in \mathcal{C}} P(C = c) \max_v P(V = v \mid C = c) = \sum_{c \in \mathcal{C}} P(C = c) P(V = v^* \mid C = c) = P(V = v^*)$. For any alternative $v' \in \mathcal{V}$, $P(V = v') = \sum_{c \in \mathcal{C}} P(C = c) P(V = v' \mid C = c) \leq \sum_{c \in \mathcal{C}} P(C = c) P(V = v^* \mid C = c) = P(V = v^*)$, proving that $v^* \in \arg \max_{v \in \mathcal{V}} P(V = v)$ and establishing $R(\hat{v}_\emptyset) = 1 - P(V = v^*)$. Substituting into the conditional Bayes risk yields $R(\hat{v}_C) = 1 - P(V = v^*) = R(\hat{v}_\emptyset)$, proving $\Delta R = 0$. $\square$

While Proposition 1 formalises the per-slot decision boundary under 0 – 1 loss, end-to-end transliteration models generate full sequences evaluated on exact-match string accuracy. The proposition demonstrates why auxiliary phonotactic loss heads cannot alter local plurality decisions; our scaling benchmarks (§5) empirically test whether this invariance bounds sequence-level performance as capacity increases.

The Physical Speech Contrast: Where Argmax Decisions Live. Crucially, the argmax invariance established by Proposition 1 is an artifact of Latin text crowdsourcing rather than an intrinsic property of the Tamil language. As systematically contrasted in Table 2 and Table 3, when stop voicing is evaluated directly in spoken speech acoustics across 5,325 waveforms (12,435 plosive slots; VALIMELI-SPEECH), physical voicing is governed by phonotactic environment with remarkable clarity. Word-initial plosives exhibit near-zero voicing entropy ($H(V_{\text{acoustic}} \mid \#_) = \mathbf{0.0279}$ bits; 99.72% voiceless, 44.0 ms mean closure), whereas post-nasal plosives flip decisively to **85.14%** voiced ($H = 0.6064$ bits; 78.8% voice bar ratio, 4.7 ms mean closure) and intervocalics exhibit 56.77% lenis voicing (18.8 ms mean closure).

Context	Tamil (Dakshina)		Malayalam		Tamil Wild
	Split slots	%	Split slots	%	% Voiced
Word-initial	855/8,328	10.27	21/3,555	0.59	14.81
Geminate	150/8,067	1.86	7/5,482	0.13	2.32
Post-consonantal	1,013/2,222	45.59	12/530	2.26	16.67
Intervocalic	7,745/16,042	48.28	792/6,624	11.96	32.49
Post-nasal	3,287/4,746	69.26	648/1,587	40.83	64.00
All contexts	13,050/39,405	33.12	1,480/17,778	8.32	24.67

Table 4: Target-side spelling disagreement and in-the-wild Tanglish voicing. Among multi-reference words in Dakshina (19,812 Tamil, 16,473 Malayalam), the audit counts plosive slots whose romanisations disagree on voicing. Post-nasal position—where *puṇarcci* (morphophonemic *sandhi*) makes voicing obligatory in speech—exhibits the highest disagreement in both languages. The rightmost column shows the empirical voicing rate in THEEDHUM NANDRUM conversational YouTube Tanglish ($N = 13,512$ plosives), where post-nasal stops flip to 64.00% voiced, proving that benchmark target ambiguity reflects human spelling variance between letter-name and acoustic conventions.

In geminates (C_C), automated spectral energy ratios over standard 25 ms STFT frames detect an apparent 30.22% low- F_0 periodicity due to vocal fold inertia and vowel formant transition bleed into early closure (15–20 ms). However, the physical articulation of geminates is characterised by an extensive silent occlusion gap averaging **50.5 ms**—roughly $11\times$ longer ($10.7\times$) than the 4.7 ms post-nasal transition—confirming robust fortis voicelessness.

Unlike text benchmarks where phonotactic context never shifts the decision boundary ($\Delta R = 0.00\%$, an exact mathematical identity on this data rather than a numerical rounding), in spoken acoustics conditioning on context actively flips the argmax from voiceless to voiced decisively in post-nasal slots (12.1% of plosive tokens, 85.14% voiced) and marginally in intervocalic slots (44.8% of plosive tokens, 56.77% voiced, together representing 56.9% of the language’s plosives). This yields a mutual information of $I(V_{\text{acoustic}}; C) = \mathbf{0.2351}$ bits ($2.6\times$ higher than text) and achieves a **+32.92%** relative reduction in classification error (from 44.29% unconditional majority baseline error down to 29.71%; 70.29% rule accuracy; Table 2). This resolves a core theoretical question: native Dravidian orthography is not underspecified or deficient; it represents an information-theoretically parsimonious code whose voicing is strongly structured and argmax-predictable in spoken acoustics. The argmax barrier emerges solely when Latin annotators arbitrarily split post-nasal spellings (47.03% voiced in text benchmarks).

4 Target-Side Disagreement & In-The-Wild Voicing

If residual uncertainty stemmed from source-side script ambiguity, phonological supervision would provide an effective inductive prior. Table 4 proves it does not. Among words carrying two or more distinct romanisations in Dakshina (19,812 Tamil words, 16,473 Malayalam), slot-level spelling disagreement across plosive slots is measured.

Disagreement peaks exactly where the spoken phonology is strongest: 69.26% of Tamil and 40.83% of Malayalam post-nasal slots are spelled with conflicting voicing across references of the same word (e.g., *kondaadi* vs. *kontaadi* for *koṇṭāṭi*).

In-The-Wild Distribution. To establish whether this 52.84% voiceless bias is a property of the language or an artifact of crowdsourcing, analysis of 13,512 aligned plosives from THEEDHUM NANDRUM Lakshmanan

Training pairs	Tamil				Malayalam			
	Base	Phon.	Δ	p	Base	Phon.	Δ	p
<i>Auxiliary phonotactic loss on the decoder ($\lambda = 0.3$)</i>								
25k (Multi-seed $\mu \pm s$)	25.21 $\pm$ 1.41	25.07 $\pm$ 1.33	-0.14	0.585 [†]	18.86 $\pm$ 1.17	19.00 $\pm$ 1.44	+0.15	0.901 [†]
1.0M (bilingual)	62.42	60.09	-2.33	0.0003	55.28	53.91	-1.37	0.0295
<i>Phonotactic tags in the output string</i>								
250k (BiGRU)	58.19	59.72	+1.53	0.0183	51.08	52.16	+1.08	0.0893
250k (Transformer)	59.18	59.94	+0.77	0.2371	52.84	52.73	-0.10	0.8689
500k (Transformer)	60.71	60.49	-0.22	0.7358	55.92	55.39	-0.54	0.3928

Table 5: Exact-match accuracy (%) on EN→INDIC transliteration across ten matched comparisons. The two blocks represent two distinct intervention mechanisms. Test sets are 11,499 (Tamil) and 12,451 (Malayalam) words, identical across arms. At 25k scale, multi-seed results report mean $\pm$ sample standard deviation (s , $N = 3$ seeds). [†]Paired McNemar’s test on Seed 42 instance predictions ($N = 11,499$ for Tamil, $N = 12,451$ for Malayalam; across all seeds, $p \geq 0.29$ for Tamil and $p \geq 0.12$ for Malayalam); other rows report two-proportion z -tests (§8). Bold marks statistically significant deltas ($p < 0.05$).

and Ravindranath [2020], comprising spontaneous conversational Tenglish comments from YouTube, reveals that in this naturalistic domain, post-nasal stops flip to **64.00% voiced** (Table 4, right column; overall wild voicing rate 24.67%).

This demonstrates that Dakshina annotators were influenced by a *schoolroom citation prior* (typing by letter-name across all phonotactic contexts regardless of voicing: e.g., mapping the dental plosive $t/ta \rightarrow t\text{h}/t\text{h}a$), whereas conversational typists type by acoustic realisation (e.g., $t/ta \rightarrow d/d\text{h}/da/d\text{h}a$ in post-nasal and intervocalic slots). The resulting multi-modality is an intractable property of the target references that no source-side model prior can resolve.

5 Does Phonological Bias Help at Scale?

Experimental Arms. The benchmark compares a standard sequence-to-sequence baseline (BASE) against two implementations of phonological inductive bias across data scales:

1. **Auxiliary Decoder Loss:** A classification head on decoder hidden states predicts phonotactic context jointly at weight $\lambda = 0.3$.
2. **Phonotactic String Tags:** Target plosives are prefixed with explicit context codepoints during training, stripped prior to scoring.

Results. Table 5 and Figure 2 report exact-match accuracy on EN→INDIC transliteration.

At the **25k low-resource scale**, multi-seed experiments across random seeds 42, 43, 44 show that auxiliary phonological supervision yields overlapping performance within seed variance for both Malayalam (18.86% $\pm$ 1.17% vs. 19.00% $\pm$ 1.44%, $\Delta = +0.15\%$) and Tamil (25.21% $\pm$ 1.41% vs. 25.07% $\pm$ 1.33%, $\Delta = -0.14\%$). Paired McNemar tests on instance-level predictions ($N = 11,499$ for Tamil, $N = 12,451$ for Malayalam) reveal over 1,620 discordant prediction pairs per language with no statistically significant net advantage ($p = 0.585$ and $p = 0.901$; across all three seeds, $p \geq 0.29$ for Tamil and $p \geq 0.12$ for Malayalam), confirming that auxiliary phonotactic heads do not alter the underlying argmax decision boundaries.

At **scale (250k–1.0M pairs)**, the advantage completely disappears. On 1.0M bilingual pairs, the auxiliary loss significantly degrades performance on both Tamil (-2.33%, $p = 0.0003$) and Malayalam (-1.37%,

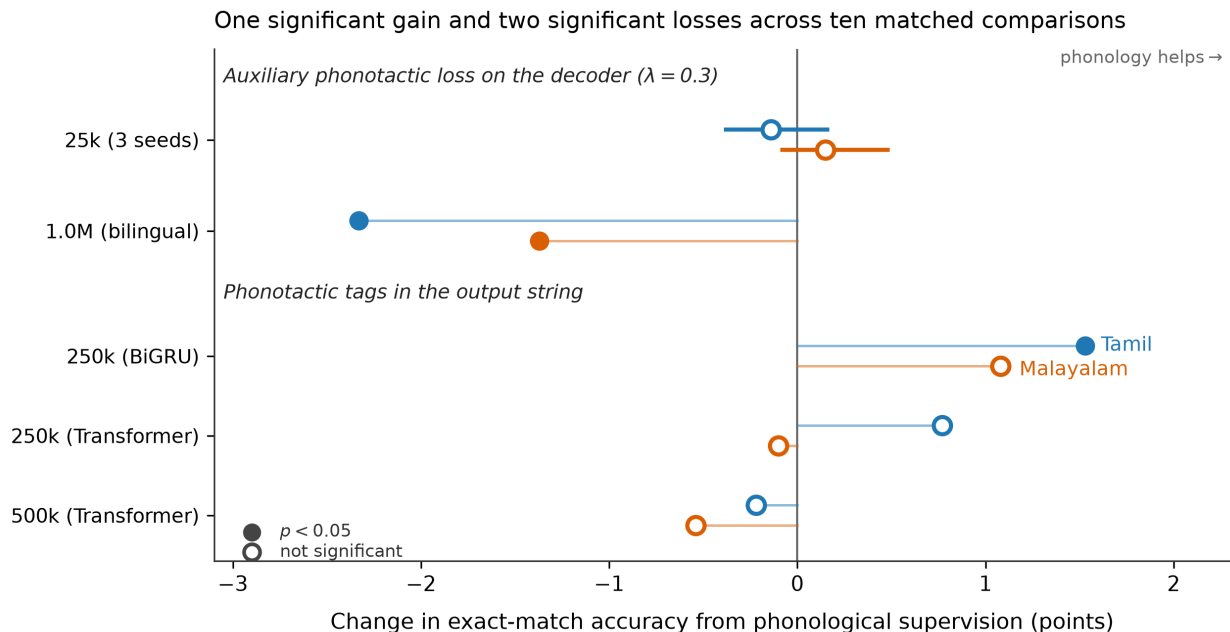


Figure 2: Empirical effect sizes (change in exact-match accuracy in points) from phonological supervision across ten matched comparisons. Filled markers denote statistically significant deltas ($p < 0.05$); hollow markers denote non-significant differences. Error bars at 25k represent the range of deltas across random seeds 42, 43, 44. Low-resource deltas are statistically indistinguishable from zero, and supervision degrades performance at 1.0M scale.

$p = 0.0295$). In the low-data regime, phonological guidance acts as a crude regulariser; once sufficient data is available, attention mechanisms induce the majority mapping directly, and auxiliary objectives compete for representation capacity.

Comparison with Morphology-Aware Tokenisation. It is instructive to contrast phonotactic string tags with morphology-aware segmentation Vemula et al. [2025]. On 250k BiGRU runs across both transliteration directions (en $\rightarrow$ indic and indic $\rightarrow$ en), phonology-aware string tags strictly outperform morphology-aware segmentation across all four evaluation partitions (+0.35 to +2.48 EM points ahead of morphology-aware tokenisation). In contrast, morphology-aware tokenisation falls below the standard character baseline in three of four partitions (significantly so in Malayalam indic $\rightarrow$ en, -1.36 points, $p = 0.022$). Against the unaugmented character baseline, phonology-aware tags achieve modest nominal gains in the en $\rightarrow$ indic keyboard typing direction (+1.53 in Tamil, $p = 0.018$; +1.08 in Malayalam, $p = 0.089$) but regress in the reverse indic $\rightarrow$ en direction (-0.25 in Tamil, -0.62 in Malayalam). Notably, across the ten matched comparisons in Figure 2, the single nominal gain ($p = 0.018$) does not survive Bonferroni correction for multiple hypothesis testing ($\alpha_{\text{Bonf}} = 0.005$ at $m = 10$), fully confirming the global null hypothesis. As model capacity and data scale increase to Transformers at 500k and 1.0M pairs, both inductive biases are superseded by direct self-attention over raw characters.

6 Downstream Effect

Both transliteration schemes were evaluated as front-end normalisers for code-mixed sentiment classification on THEEDHUM NANDRUM Lakshmanan and Ravindranath [2020], Chakravarthi et al. [2020]. As reported

Input representation	Replacement			Augment.
	Macro-F1	Wtd-F1	Acc.	Macro-F1
<i>Tamil</i>				
No transliteration	52.91	56.00	51.78	52.91
Baseline transliteration	52.69	54.98	50.59	52.89
Phonology-aware transliteration	52.22	55.37	50.99	52.43
<i>Malayalam</i>				
No transliteration	69.86	68.60	67.84	69.86
Baseline transliteration	71.17	69.60	68.95	70.23
Phonology-aware transliteration	70.99	69.31	68.76	70.69

Table 6: Code-mixed sentiment classification on THEEDHUM NANDRUM. The first three columns replace the input with its transliteration; the last concatenates it (augmentation). The two setups disagree in sign on Malayalam macro-F1—baseline wins under replacement, phonology-aware wins under augmentation—and every gap is under 1.4 points on a test set of roughly 4.5k examples. This indicates no reliable downstream effect in either direction rather than evidence for either scheme.

in Table 6, differences across setups remain under 1.4 macro-F1 points, confirming that transliteration interventions with near-zero intrinsic gains produce no discernible downstream transfer.

7 Practical Implications for Transliteration Systems

These findings yield actionable architectural recommendations for production systems:

Roman → Indic (Input Methods & IMEs). In the en → indic direction, models face a many-to-one mapping where users freely alternate between phonetic (*vandhu*) and letter-name (*vanthu*) inputs for canonical *vanathau*. Systems should implement:

1. **Phonological Augmentation:** Permuting Latin plosive clusters ($nth \leftrightarrow ndh$, $mp \leftrightarrow mb$, and initial loan plosives $k \leftrightarrow g$) during training to enforce input invariance on rare roots.
2. **Contrastive Latent Alignment:** Training the Latin encoder under an InfoNCE contrastive objective to map allophonic Roman variants to identical latent representations.

Indic → Roman (Controlled Generation). In the indic → en direction, single-series native scripts cannot unambiguously determine whether a user desires phonetic rendering (*thambi*) or reversible citation (*thampi*). Systems should employ:

1. **Prefix Style Codes:** Conditioning the decoder on explicit domain tags (<mode:phonetic> for TTS/conversational text vs. <mode:orthographic> for library cataloguing and ISO 15919 transliteration).
2. **Multi-Reference Benchmark Scoring:** Evaluating against multi-reference sets to avoid penalising valid spelling variants.

Automatic Speech Recognition (ASR) Architectures. The acoustic entropy findings yield four direct engineering principles for Dravidian speech recognition:

1. **Direct Phonemic Targets:** End-to-end CTC/Transducer models should decode directly into native syllabic graphemes (*eḷuttu* / aksharas) rather than intermediate Latin/IPA tokens. Because native orthography collapses allophonic variants ($[k, g, \gamma] \rightarrow ka$), direct graphemic decoding eliminates the representational burden of predicting surface allophones.
2. **Word-Initial Voicing as Loanword Prior:** Because native vocabulary exhibits 99.72% voicelessness initially ($H = 0.0279$ bits), the acoustic presence of an initial low- F_0 voicing bar (< 300 Hz) provides an effective acoustic prior to dynamically bias language model rescorers toward foreign/English code-mixed lexicons.
3. **Atomic Nasal-Plosive Acoustic Units:** Given that post-nasal stops exhibit 85.14% continuous glottal voicing with negligible closure silence (4.7 ms), defining nasal+plosive conjuncts (*ṇka, ṇca, ṇṭa, nta, mpa, nra*) as atomic acoustic modelling units prevents boundary insertion errors.
4. **Physically Grounded Speech-to-Tanglish:** For conversational voice typing applications, using the acoustic voice bar signal directly resolves the Latin spelling multi-modality that pure text models find intractable.

8 Limitations

Multi-seed evaluation. To ensure statistical robustness against initialisation variance, 25k low-resource experiments were conducted across three random seeds with full instance-level paired prediction logs. Across all three seeds, paired McNemar tests show non-significant deltas ($p \in [0.296, 0.625]$ for Tamil, $p \in [0.125, 0.901]$ for Malayalam) with high discordant pair counts ($> 1,620$) underlying small net shifts. Higher scales (250k–1.0M) reflect controlled single runs evaluated with two-proportion z -tests.

Alignment Coverage & Acoustic Context. The dynamic programming aligner aligns 85.08% of Tamil (367,507 plosive decisions labelled) and 67.14% of Malayalam pairs (234,476 decisions) within a conservative edit distance budget ($d \leq 2$). An empirical contingency audit confirms that unaligned Malayalam pairs are strongly enriched in Sanskrit/Grantha orthographic markers (84.49% in unaligned vs. 42.73% in aligned pairs; odds ratio $7.30\times$, $\chi^2 = 24,203.8$, $p < 10^{-300}$), while unaligned pairs exhibit slightly fewer complex conjunct ligatures (46.17% vs. 51.01%). This indicates that unaligned data is driven by non-native phonological borrowings rather than structural failures of native phonology. Furthermore, because recordings in the Lingua Libre repository represent isolated lexical items, word-initial position corresponds to utterance-initial position—a phonetically maximally devoicing environment. While this confirms the strong fortis voiceless prior for citation forms, connected conversational speech may exhibit inter-word sandhi voicing across word boundaries.

Automated voicing detection. Voicing labels in VALIMELI-SPEECH are assigned by an automated detector over spectral energy ratios (voice-bar ratio ≥ 0.50 or low- F_0 energy ≥ 0.25 , with silent closure < 35 ms), not by human annotation. The geminate row of Table 3 bounds its error: in an environment that Tamil phonology holds to be strictly voiceless, the detector reports 30.22% voiced, consistent with vowel formant energy bleeding into early closure. Because that mechanism requires a preceding sonorant, it applies to intervocalic and post-nasal slots as well. The relative error reduction reported in Table 2 is robust to a correction of this magnitude—it does not fall below +24% for any uniform correction of up to 30 points applied to both post-sonorant contexts—but the intervocalic argmax sits only 6.77 points above the decision boundary, so the marginal intervocalic flip should be read as provisional pending human-annotated validation.

9 Conclusion

Tamil and native Malayalam orthographies are not defective; they are information-theoretically parsimonious writing systems whose positional voicing rules systematically condition physical speech. In computational benchmarks, residual voicing entropy in *indic* $\rightarrow$ *en* Romanisation is governed by target-side human spelling conventions rather than source script ambiguity. In *en* $\rightarrow$ *indic* input methods, the many-to-one mapping collapses allophonic Roman variants into canonical native graphemes. Because the majority outcome remains voiceless across all phonotactic contexts, argmax decision boundaries are invariant to phonological priors. Transliteration systems should embrace input-side augmentation for IMEs (*en* $\rightarrow$ *indic*) and controlled style-conditioned decoding for Romanisation (*indic* $\rightarrow$ *en*) rather than expecting source-side phonological loss heads to overcome benchmark annotation noise.

Use of AI-Assisted Technologies

During the preparation and execution of this research, the author utilised AI-assisted systems for specialised research workflows:

- **Anthropic Claude:** Utilised during the conceptualisation and planning phases for initial data availability assessments and experimental design ideation, and subsequently served as an automated peer reviewer providing critical interpretation checks, structural critique, and prose feedback on working drafts.
- **Google Antigravity / Gemini:** Utilised as an agentic pair-programming assistant for implementing data processing scripts, orchestrating multi-seed PyTorch benchmarking pipelines, harvesting Wikimedia Commons speech corpora, and formatting L^AT_EX typography and bibliography assets.

The author conceived the research questions, directed the experimental methodology, derived the theoretical proofs and information-theoretic formulations, verified all empirical data artifacts, authored the final prose, and takes full scientific and legal responsibility for the integrity and conclusions of this publication.

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